

# EVALUATING ROBUSTNESS OF A HMM-BASED CLASSIFICATION SYSTEM OF VOLCANO-SEISMIC EVENTS AT COLIMA AND POPOCATEPETL VOLCANOES

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## ABSTRACT

This work presents a continuous volcano-seismic classification system based in the Hidden Markov Models as solution to recently strong needs for automatic event detection and recognition methods in early warning and monitoring scenarios. Furthermore, our system includes a reliable method to assign confidence measures to the recognized signals in order to evaluate the robustness of the results. Data from the two most active volcanoes have been used to probe the system reliability on a complex joint corpus achieving a recognition accuracy higher than 78% in blind recognition tests.

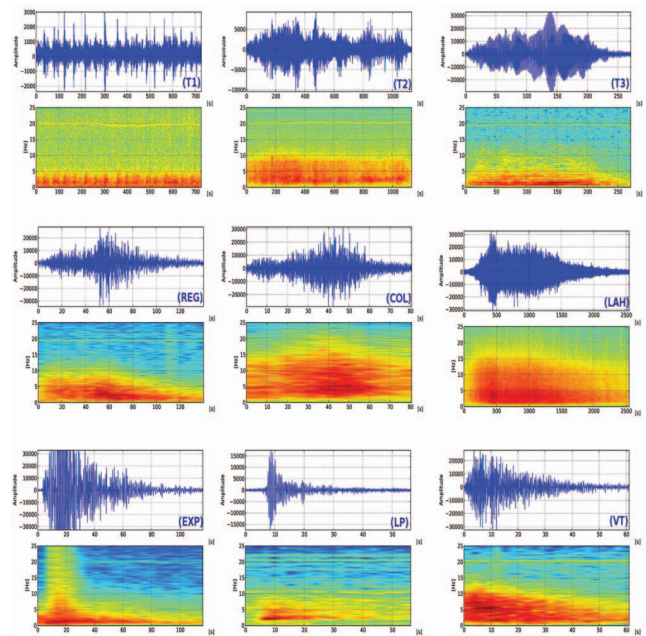
**Index Terms**— volcano monitoring, classification, Hidden Markov Models, volcano-seismic events, reliability

## 1. INTRODUCTION

The event classification at active volcanoes is usually a hard and time-consuming task carried out by expert technicians in a non-stop monitoring process. Nowadays, automatic classification and detection methods are being developed in order to help with this task and for building early warning robust systems. In this area, the Hidden Markov Models (HMM) [1, 2] are arising as the most promising solution as it was in Automatic Speech Recognition (ASR) [3, 4, 5, 6] versus other approaches [7]. The HMM approach is also valid when SNR is too low for the classic detection algorithms to properly work [6].

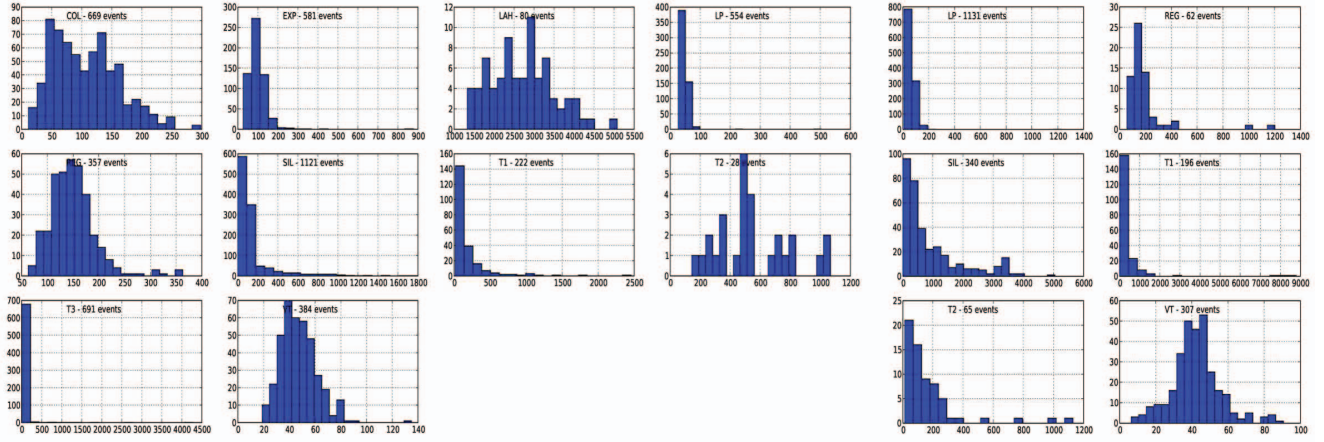
In the present paper we advance one step forward following the ideas already stated in ASR. Previous works [3, 4, 6] have probed the parallelism between speech and volcano-seismic events in terms of signal complexity and real-time requirements. Multi-speaker ASR databases lead to independent classification models able to transcribe the speech of an unknown speaker. Our aim is the design of a general recognition system integrating several volcano-seismic classes from different types of volcanoes to be effortless portable to any

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**Fig. 1.** Seismograms and spectrograms (frequency versus time) of the database classes: (T1) tremor with repetitive pulses, (T2) spasmodic tremor, (T3) harmonic tremor, (REG) regional earthquakes, (COL) collapses, (LAH) lahars, (EXP) explosions, (LP) long period events and (VT) volcano-tectonic earthquakes.

volcano monitoring station. At first approach, events from two Mexican volcanoes of the same type have been mixed for building a complex database used to evaluate the robustness of a continuous HMM-based recognition system. For increasing system reliability, a method to assign class confidence scores to each recognized event has been implemented facilitating result interpretation by experts in a critical alert situation or simply in a signal cataloging or monitoring work. We also made a comparative study between the system of the mixed corpus and the ones built with standalone databases



**Fig. 2.** Time distribution histograms (y-axis: nº of events / x-axis: event duration [s]) for each volcano-seismic class of Colima (the four left columns) and Popocatepetl (the two right columns).

analyzing independent results for each event class.

## 2. DATABASE DESCRIPTION

In order to build a solid database the two most active Mexican volcanoes were selected. Popocatepetl and Colima are andesitic stratovolcanoes, so they share event generation sources and thus, the same type of events [8]. The Volcán de Fuego sited at Colima city has erupted more than 40 times since 1576. The largest eruption in recent years occurred in 2005. The dataset was collected at 2 monitoring stations of short period plus one broad band station mainly during 2004-2006 period and with some events from 1998. Popocatepetl counts with more than 20 eruptions reported alongside its history, occurring the most violent in December 2000. The events were collected between November and December of 2002 at a short period station.

The data were manually labeled by 3 expert technicians into 7 classes (see Figure 1): explosions (EXP), long period (LP) events, collapses (COL), lahars (LAH), regional tectonic (REG) and volcano-tectonic (VT) earthquakes, seismic noise (SIL) and 3 subclasses of volcanic tremor; with repetitive pulses (T1), spasmodic (T2) and harmonic (T3). The Colima dataset achieves more than 195 hours registered in 2590 files and a total of 4687 events (1121 SIL, 357 REG, 222 T1, 28 T2, 691 T3, 669 COL, 581 EXP, 80 LAH, 554 LP and 384 VT) while Popocatepetl has almost 139 hours distributed in 680 recordings identifying 2101 events (1131 LP, 62 REG, 340 SIL, 196 T1, 65 T2 and 307 VT). Figure 2 shows the time distribution histograms of these class events. A joint database including the events from Colima and Popocatepetl was also built in order to test system robustness in a complex classification scenario.

## 3. BUILDING THE CHMM-BASED AUTOMATIC CLASSIFICATION SYSTEM

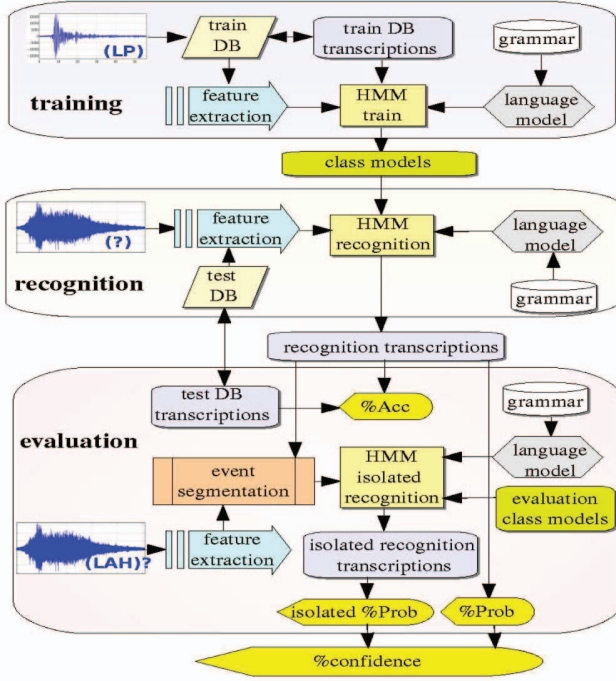
Automatic continuous classification systems based in HMM state of the art has been built by Colima and Popocatepetl corpus. A speech-based Mel Frequency Cepstral Coefficients (MFCC) parametrization of seismic data is achieved to extract relevant signal features [3, 4] instead of classic seismic properties used in previous works [6, 7]. Multivariate Gaussian probability density functions will capture the underlying statistics of the feature space during successive training iterations of the models.

A flexible left to right HMM topology has been adopted named as multi-state HMM which allows to have different number of states for each class model. This number has to be proportional to the class time and feature space variability. Initial values can be set observing the event time distribution (Figure 2) of the classes and their spectrograms (Figure 1) because MFCC features are based on the Short Time Fourier Transform (STFT) and signal smoothing performed in the spectrogram calculation.

The Figure 3 represents the basic scheme of our classification system. Often, several iterations over training, recognition and evaluation stages are required for finding optimal configuration values to build the best class models. The language model takes account of contextual event information applying statistics and/or deterministic rules over the sequence of recognized events. In our case it is used only to discriminate between continuous or isolated evaluation tests, being the same in the 3 stages.

## 4. SYSTEM EVALUATION

In order to set-up the system and check database reliability, we perform continuous close and blind isolated event recognition tests: close tests use the same data partition to train the



**Fig. 3.** Stages involved in the HMM-based automatic classification system: a) building and training the class models, b) recognition of data and c) evaluation of the system.

models and to evaluate them while in blind tests the training data partition is not the used for evaluation. In continuous recognition the data files can contain an arbitrary number of events per file but in isolated event recognition only one event per file is allowed and the recognizer is forced to not commit insertion (*I*) (detection of more than the existing events in one data file) or deletion (*D*) (not detection of some event) errors. The accuracy (%Acc) is the most important flag in recognition scores and is computed by:

$$\%Acc = 100 \frac{H - I}{N} = 100 \frac{N - D - S - I}{N} \quad (1)$$

being (*H*) the events successfully classified and (*S*) the wrong labeled of a total of (*H*) events in the evaluation partition. The equation (1) reveals us the insertion (*I*) negative influence in %Acc. The accuracy will be low at first recognition iterations to later increase as much as the number of output distribution components is increasing too. Once the maximal %Acc is reached at a certain iteration, it will decrease in a smoothed logarithmic oscillating way as the Gaussian components increase. Indeed, the higher components are used the higher decoding computation time is required, so few Gaussians (from 1 to 20) are encouraged to use.

#### 4.1. Result reliability: class confidence scores

The result reliability is usually performed assigning a confidence score to each recognized event based in the maximum

output likelihood computed by the class models. A goal of our classification system is the ability to assign as many confidence measures as classes in the database, giving for each event the probability of matching each class model in % percent. The method (Figure 3) segments the events of the evaluation partition according to the time labels of the recognition transcriptions for performing an isolated classification over the segmented data. An isolated recognition process is repeated using each time only one class evaluation model to obtain the output likelihood of this class in the transcriptions. Finally, all the class probabilities are normalized by the probability values of the recognition transcriptions for achieving confidence scores.

#### 4.2. Evaluation results

To improve the reliability of the results and minimize the negative effect that the lack of events has in the HMM, blind recognition is carried out averaging the results of 3 tests where alternatively 2 partitions are used to train models and the other is for evaluation. All models built from the same database have the same Gaussian components in order to establish a direct comparison between the continuous close, isolated blind and continuous blind results, using 12, 15 and 12 components for Popocatepetl, Colima and the joint corpus respectively.

| CONTINUOUS CLOSE - (%)class accuracy |       |       |       |       |        |       |       |       |       |       |
|--------------------------------------|-------|-------|-------|-------|--------|-------|-------|-------|-------|-------|
|                                      | SIL   | EXP   | LP    | REG   | T1     | T2    | T3    | COL   | LAH   | VT    |
| Popo                                 | 58.75 |       | 84.90 | 87.10 | 69.90  | 73.85 |       |       |       | 93.14 |
| Colima                               | 83.59 | 96.39 | 94.95 | 97.2  | 81.53  | 89.29 | 80.90 | 84.01 | 74.68 | 92.45 |
| Joint                                | 77.14 | 96.04 | 88.30 | 90.45 | 65.07  | 66.67 | 77.42 | 82.21 | 75.64 | 91.88 |
|                                      |       |       |       |       |        |       |       |       |       |       |
|                                      |       |       | Popo  |       | Colima |       | Joint |       | MEAN  |       |
| (%)mean class accuracy               |       |       | 77.94 |       | 87.50  |       | 81.08 |       | 82.17 |       |
| (%)event accuracy                    |       |       | 80.21 |       | 87.73  |       | 83.48 |       | 83.81 |       |
|                                      |       |       |       |       |        |       |       |       |       |       |
| ISOLATED BLIND - (%)class accuracy   |       |       |       |       |        |       |       |       |       |       |
|                                      | SIL   | EXP   | LP    | REG   | T1     | T2    | T3    | COL   | LAH   | VT    |
| Popo                                 | 97.85 |       | 93.56 | 79.13 | 84.97  | 23.49 |       |       |       | 89.30 |
| Colima                               | 82.91 | 95.53 | 90.63 | 93.28 | 79.73  | 59.63 | 84.41 | 91.96 | 96.15 | 87.24 |
| Joint                                | 91.92 | 96.56 | 86.6  | 90.89 | 81.09  | 49.53 | 86.16 | 92.09 | 94.87 | 85.04 |
|                                      |       |       |       |       |        |       |       |       |       |       |
|                                      |       |       | Popo  |       | Colima |       | Joint |       | MEAN  |       |
| (%)mean class accuracy               |       |       | 78.05 |       | 86.15  |       | 85.48 |       | 83.22 |       |
| (%)event accuracy                    |       |       | 90.24 |       | 87.99  |       | 88.66 |       | 88.96 |       |

**Table 1.** Results for continuous close and isolated blind classification.

The table 1 lists the results for the continuous close and isolated blind tests, the so-called *integrity tests*, which are useful for system initialization and setup. Event accuracy scores 80.21%, 87.73% 83.48% in close experiments for Popocatepetl, Colima and the mixed data. Isolated recognition was also performed to analyze the system response without event insertion and deletion issues reaching 90.24%, 87.99% and



| (% )class accuracy     |       |       |       |        |       |       |       |       |       |       |
|------------------------|-------|-------|-------|--------|-------|-------|-------|-------|-------|-------|
|                        | SIL   | EXP   | LP    | REG    | T1    | T2    | T3    | COL   | LAH   | VT    |
| Popo                   | 55.18 |       | 77.32 | 62.88  | 58.21 | 13.87 |       |       |       | 86.21 |
| Colima                 | 85.17 | 90.04 | 88.02 | 91.58  | 58.09 | 41.67 | 70.39 | 78.29 | 64.97 | 86.98 |
| Joint                  | 76.11 | 93.12 | 83.66 | 84.01  | 53.6  | 19.19 | 69.44 | 77.27 | 59.41 | 88.98 |
|                        |       |       |       |        |       |       |       |       |       |       |
|                        | Popo  |       |       | Colima |       |       | Joint |       | MEAN  |       |
| (%)mean class accuracy | 58.95 |       |       | 75.52  |       |       | 70.48 |       | 68.32 |       |
| (%)event accuracy      | 72.42 |       |       | 81.71  |       |       | 78.29 |       | 77.47 |       |

**Table 2.** Results for continuous blind classification.

88.66%. The tables also include class accuracy values, useful for monitoring and discriminate class results and mean scores for a general perspective of the recognition process.

Finally, blind tests were carried out achieving 72.42%, 81.71% and 78.29% event accuracy results for the above mentioned databases (Table 2). In order to evaluate the robustness of the result reliability assignation method a re-scoring selecting the maximal class confidence score to label each recognized event was made obtaining similar blind test rates: 72.80% for Popocatepetl, 81.78% for Colima and 78.17% for the joint database.

## 5. CONCLUSIONS

In this work we have implemented a CHMM-based automatic classification system built by a database of 8 classes and 3 subclasses collected from 2 volcanoes. A score of 78% event accuracy in blind test is fair enough given the complexity of the recognition task.

The review of the evaluation results and the experimentation work reveals relevant information useful for improving the system. Multi-state HMM slightly overcome fixed-state models allowing a class independent compensation of inserted and deleted events in recognition, which is a way to the increase class accuracy. Continuous close tests achieve poor results (83.81% event Acc.) compared with isolated blind (88.96%) while any speech HMM-based system performing the same task should obtain more than 99%. This issue, besides the low scores for T1, T2 and LAH classes uncover the requirement of new event features to be developed. The accuracy scores in blind tests are higher when the n° of events/files (3.09, 1.81 and 2.08 for Popocatepetl, Colima and the joint corpus) is low. This is related with the difficulty to correctly segment the events in a continuous data stream that leads to deletion and insertion errors. The low accuracy for classes with high time variability as T1, T2, T3. LAH or SIL is mainly due to these errors. Nevertheless, they are not relevant from a geophysical point of view [4]. The fact that the re-scoring process, based on confidence class values, reaches similar results as blind recognition unveils that the confidence scoring system is reliable enough for cataloging purposes or for being integrated in a real-time classification

monitoring station.

It is worthwhile to note that the supervised event labeling process is the key to achieve precise models, and human technicians sometimes suffer the lack of an unified criteria which hardly can overcome a 80% of agreement in manual classification [4]. The time to replace human operators in monitoring tasks maybe has come, allowing them to paying their attention towards to result interpretation and alert situations.

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